

Installation

MicroICS is a tool to analyse 3D biofilm images. It runs as a self-contained Docker image (<https://hub.docker.com/r/skblaz/microics>), accessed through a browser-based interface, and the source code is available on GitHub (<https://github.com/SkBlaz/biofilm-classification>).

Installation on Windows

● Install Docker Desktop

- On the Docker download page (<https://docs.docker.com/desktop/setup/install/windows-install/>), click the button for your version of Windows (amd64 or arm64). The download will start.

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Install Docker Desktop on Windows

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Docker Desktop terms

Commercial use of Docker Desktop in larger enterprises (more than 250 employees OR more than \$10 million USD in annual revenue) requires a [paid subscription](#).

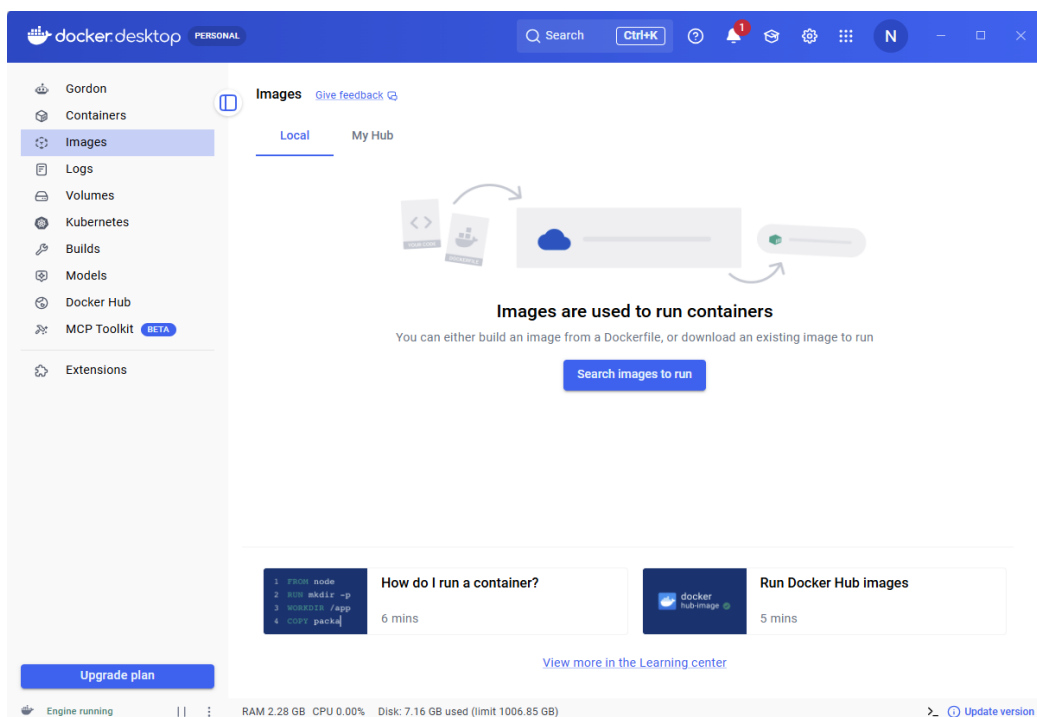
This page provides download links, system requirements, and step-by-step installation instructions for Docker Desktop on Windows.

Docker Desktop for Windows - x86_64

Docker Desktop for Windows - x86_64 on the Microsoft Store

Docker Desktop for Windows - Arm (Early Access)

- When it finishes, double-click the downloaded file to begin installation.
- Follow the installer, using the default (recommended) options.
- Once installed, start Docker Desktop and wait until the whale icon indicates that it is running. Active Docker Desktop looks like this:



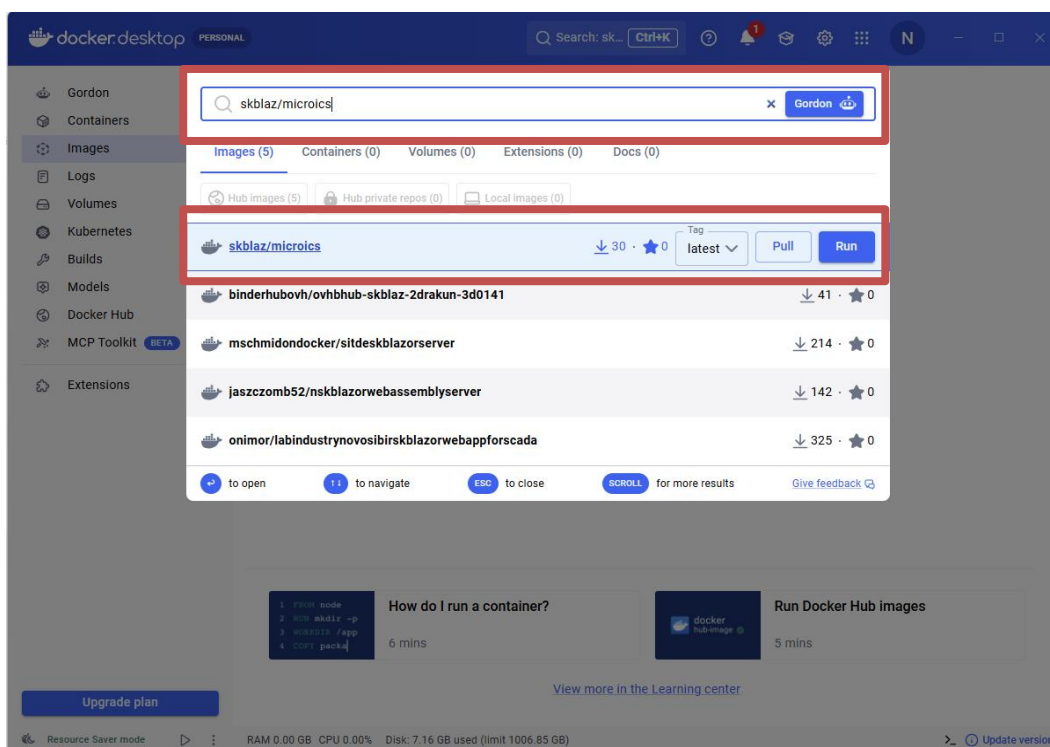
● Setting up MicroICS (via Docker)

You can do this in two ways: using Docker Desktop (the graphical interface – make sure it is running and start it if not), or using the command line (PowerShell on Windows, Terminal on macOS/Linux).

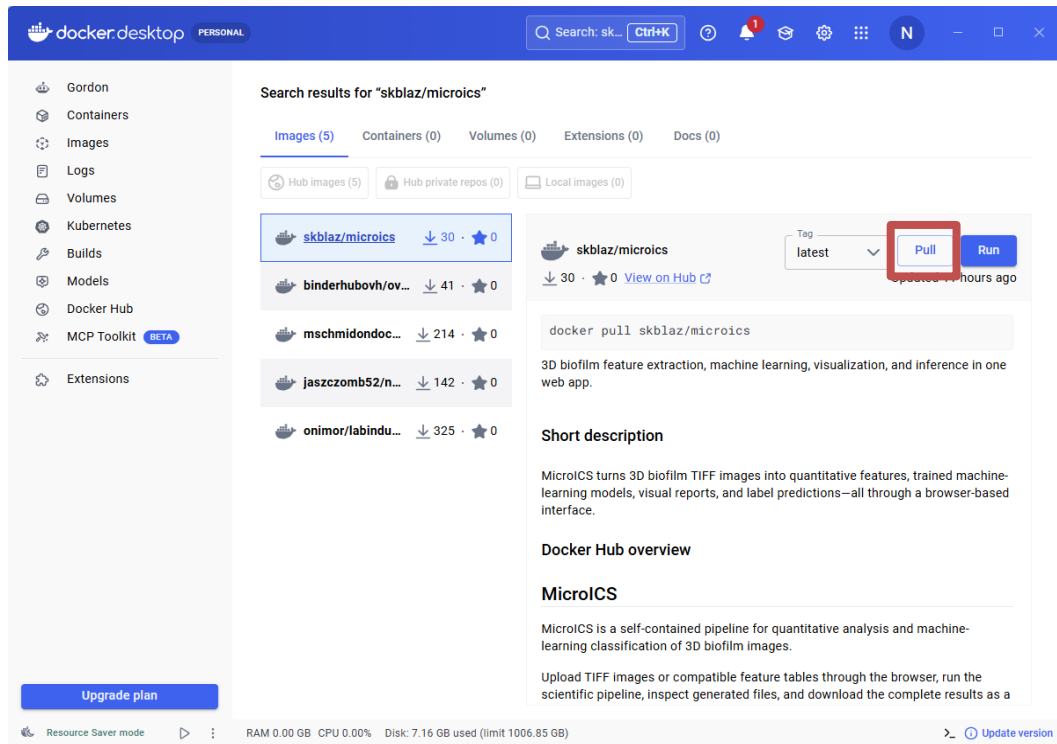
Before Docker is installed and run, WSL, the virtual machine, and Hyper-V may need to be activated on Windows. You can do this by searching for “Windows Features” and selecting “Turn Windows features on or off”. Tick Hyper-V, Virtual Machine Platform, and “Windows Subsystem for Linux”. Click OK and restart the computer, then continue installing Docker. If other issues arise, please resolve them with your IT department.

Option 1 — using Docker Desktop (graphical)

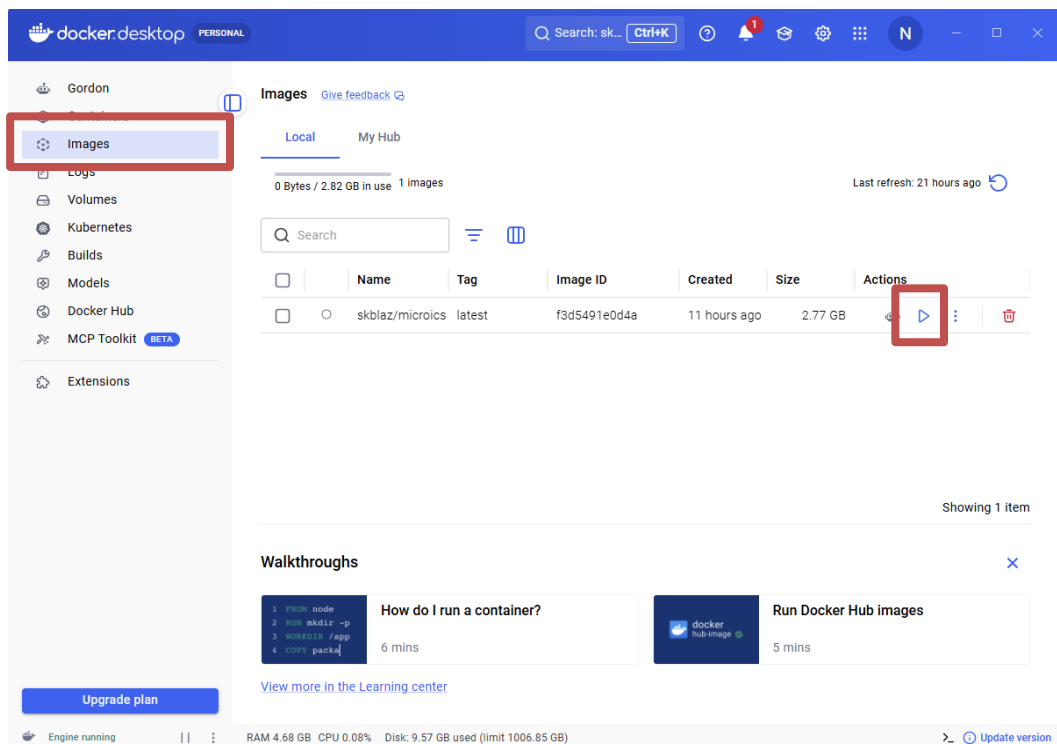
- In the search bar at the top of Docker Desktop, type **skblaz/microics**.



- Select it by clicking, then click **Pull** to download it. This may take a few minutes.

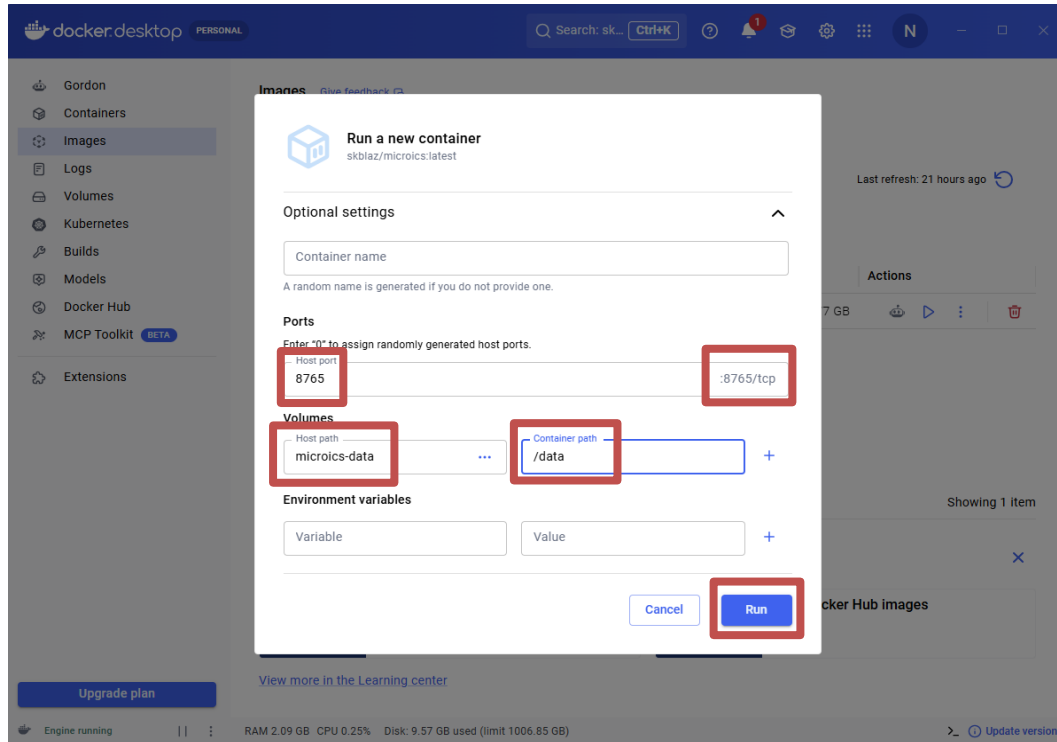


- When it has downloaded, open the **Images** tab on the left. Find skblaz/microics and click **Run** (the ► arrow on the right).

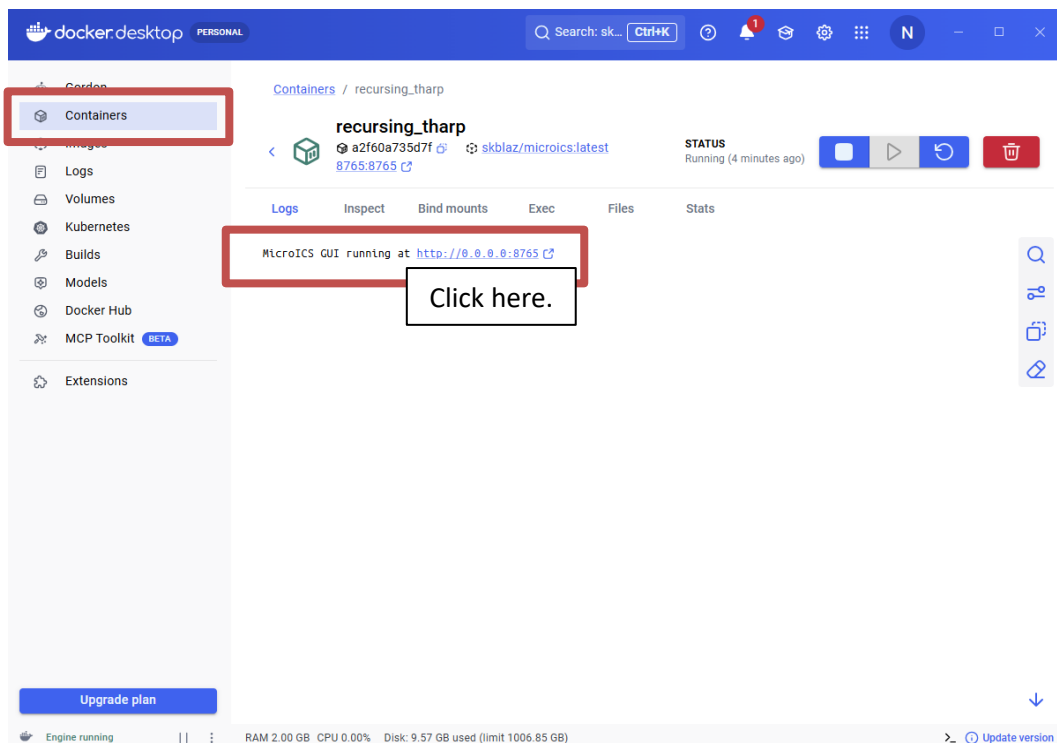


- In the dialogue that appears, click **Optional settings** to expand it, and enter:
 - **Ports** — Host port: 8765 (Container port: 8765, in grey, usually automatically filled)
 - **Host path**: microics-data, **Container path**: /data

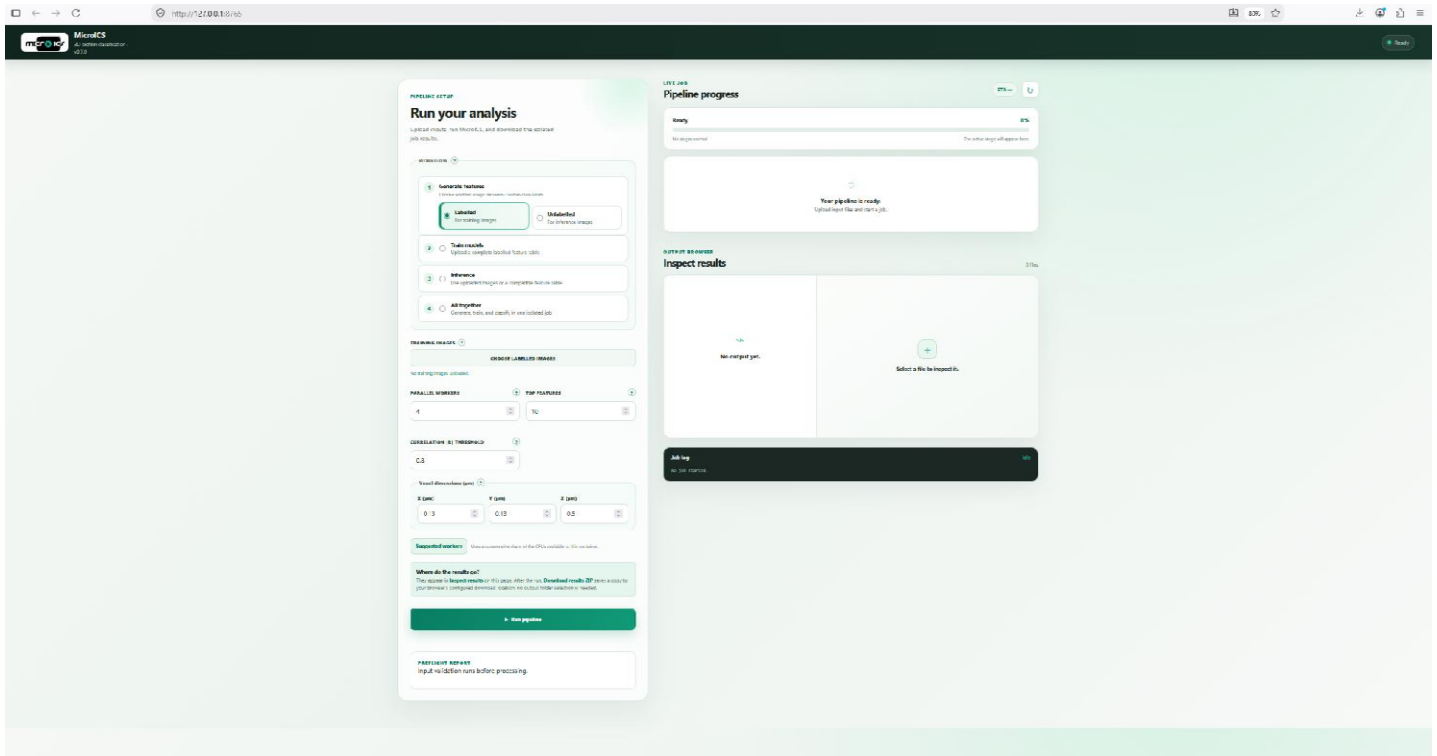
After you enter these, click **Run**.



- The container now appears under the **Containers** tab. Docker Desktop shows a clickable **8765:8765** link next to the running container.



- The MicroICS interface should look like this:



Option 2 — using the command line

- Make sure Docker Desktop is running, then open PowerShell (Windows) or Terminal (Mac/Linux).
- Download the image:

```
docker pull skblaz/microics:latest
```

- Start the tool:

```
docker run -p 8765:8765 -v microics-data:/data skblaz/microics:latest
```

- Open <http://localhost:8765> in your browser.
- To stop the container press on (■) below the Actions.

● Restarting the existing container of MicroICS

1. Start Docker Desktop and go to Containers.
2. Find your MicroICS container (it'll be stopped).
3. Click the Start button (▶) next to it.
4. Once it's running, click the 8765:8765 link, or open <http://localhost:8765> in your browser.
5. When you have finished the analyses, stop the container by pressing (■) below Actions.

To get the latest version of the Docker image of MicroICS, you need to repeat the steps for setting up MicroICS (via Docker).

The image consists of two screenshots of the Docker Desktop interface, illustrating the steps to restart the MicroICS container.

Top Screenshot: The 'Containers' tab is selected in the left sidebar (labeled '1'). The main area shows 'No containers are running.' A table below lists containers. The container 'recurring_tharp' is shown with ID 'a2f60a735d7f' and image 'skblaz/mic'. The 'Port(s)' column shows '8765:8765'. The 'Actions' column has a 'Start' button (▶) which is highlighted with a red box and labeled '2, 3'.

Bottom Screenshot: The container 'recurring_tharp' is now running. The 'Port(s)' column shows '8765:8765' with a link icon, highlighted with a red box and labeled '4'. The 'Actions' column has a 'Stop' button (■) highlighted with a red box and labeled '5'. The status bar at the bottom shows 'Engine running' and system resources.

Installation on Linux

● Install Docker

- Download and run Docker's official installation script:

```
curl -fsSL https://get.docker.com -o get-docker.sh
```

```
sudo sh get-docker.sh
```

- Allow Docker to run without sudo:

By default, Docker requires sudo for every command, which causes permission problems with output files. Add your user to the Docker group so Docker can be run directly:

```
sudo usermod -aG docker $USER
```

- Then reboot for this change to take effect:

```
sudo reboot
```

- Check that Docker installed correctly (recommended)

```
docker --version
```

It should print a version number.

Then continue to the "Installing MicroICS with Docker" steps — from that point on, Linux is identical to the Windows (pull the image, run it, open the browser).

Installation on macOS

● Install Docker

- Download Docker Desktop from <https://docs.docker.com/desktop/setup/install/mac-install/> and install it. On Mac you may be offered an Apple chip version (M1–M4) or an Intel chip version – check which you have via → About This Mac → look at “Chip” or “Processor”. Install using the default options.
- Start Docker Desktop: Open Docker Desktop from Applications and wait until the whale icon in the menu bar shows it is running. Docker must be running whenever you use MicroICS.
- Check Docker is working (optional): Open Terminal and run:

```
docker --version
```

It should print a version number.

Then continue to the "Installing MicroICS with Docker" steps — from that point on, Mac is identical to the Windows or Linux (pull the image, run it, open the browser).

Running the MicrolCS

Generate features

1. Choose “Generate features” in mode “Labelled”. Labelled mode extracts the label from each filename, so that the model can link the features to the label during training.
2. Point the software to the folder containing your images in TIFF format, using “Choose labelled images”.
1. Set the running options. “Parallel workers” – how much of your CPU to dedicate to the analysis: use full capacity when you are not otherwise using the computer, and less if you are working on it at the same time. Suggested workers will provide optimal settings for running.
At this point, “Top features” and “Correlation |R| threshold” do not matter.
3. Adjust the voxel dimensions (the pixel size in micrometres) to match your images.
4. Click “Run pipeline”.

When the run starts, the software also checks the image naming and reports the number of classes it found.

The screenshot shows the MicrolCS web interface with five numbered callouts:

- 1** Points to the "Generate features" section, specifically the "Labelled" radio button which is selected. Below it, the text "Choose whether image filenames contain labels." is visible.
- 2** Points to the "TRAINING IMAGES" section, specifically the "CHOOSE LABELLED IMAGES" button.
- 3** Points to the "PARALLEL WORKERS" and "TOP FEATURES" sections. The "PARALLEL WORKERS" dropdown is set to 4, and the "TOP FEATURES" dropdown is set to 20.
- 4** Points to the "Voxel dimensions (µm)" section, which includes three input fields: "X (µm)" set to 0.13, "Y (µm)" set to 0.13, and "Z (µm)" set to 0.5.
- 5** Points to the "Run pipeline" button, which is a large green button with a right-pointing triangle icon.

Other visible elements include:

- The "Unlabelled" radio button under "Generate features" with the text "For inference images".
- The "Train models" section with the text "Upload a complete labelled feature table".
- The "Inference" section with the text "Use uploaded images or a compatible feature table".
- The "All together" section with the text "Generate, train, and classify in one isolated job".
- The "Suggested workers" section with the text "Uses a conservative share of the CPUs available to this container."
- The "Where do the results go?" section with the text "They appear in [Inspect results](#) on this page. After the run, [Download results ZIP](#) saves a copy to your browser's configured download location; no output folder selection is needed."
- The "Ready for a new isolated job." status message.
- The "PREFLIGHT REPORT" section with the text "Input validation runs before processing."

Train a model

1. Choose “Train models.”
2. Navigate to the folder containing calculated features (datafile.tsv) using “Choose feature table”.
3. Set the running options: “Parallel workers” – how much of your CPU to dedicate to the analysis: full capacity when you are not otherwise using the computer, and less if you are working on it at the same time. Suggested workers will provide optimal settings for running.
“Correlation threshold” – choose how many of the most important features to analyse, and the threshold for the analysis.
“Unit of replication” – specify what counts as an independent experiment (day, plate, well, or position within the well). This is important for testing the trained model correctly.
4. Choose which learner to train on your data: a single learner and choose learner (model) from dropdown menu, or all those currently available.
5. Click “Run pipeline.”

When the run starts, the software also checks the input datafile for missing values (NaN, empty cells, etc.).

The screenshot displays the 'Train a model' interface with five numbered callouts:

- 1** Points to the 'Train models' option in the left sidebar, which is selected. Below it are 'Inference' and 'All together' options.
- 2** Points to the 'COMPLETE FEATURE TABLE' section, which includes a 'CHOOSE FEATURE TABLE' button and a note: 'No feature table uploaded.'
- 3** Points to the configuration section containing:
 - 'PARALLEL WORKERS': A dropdown menu set to '4'.
 - 'TOP FEATURES': A dropdown menu set to '20'.
 - 'CORRELATION |R| THRESHOLD': A dropdown menu set to '0.8'.
 - 'UNIT OF REPLICATION': A dropdown menu set to 'Well'.
- 4** Points to the 'Learners' section, which includes:
 - 'One learner' (selected) and 'All learners' options.
 - 'LEARNER': A dropdown menu set to 'Random Forest'.
- 5** Points to the 'Run pipeline' button at the bottom, which is green and has a right-pointing arrow. Below the button is a status message: 'Ready for a new isolated job.'

Additional interface elements include a 'Suggested workers' note, a 'Where do the results go?' section with instructions on viewing and downloading results, and a 'PREFLIGHT REPORT' section at the bottom.

Inference

Inference consists of two steps: first, calculate features from the new images to obtain a list of features that the model can then use to make an inference on them, which is done in the second step.

Do step (1) — generate features from the new images:

1. Choose “Generate features” in “Unlabelled” mode. In unlabelled mode the labels are ignored because these images are the unknowns to be classified.
2. Point the software to the folder with the images for inference, using “Choose inference images”.
2. Set “Parallel workers” – how much of your CPU to dedicate to the analysis: full capacity when you are not otherwise using the computer, less if you are working on it at the same time. Suggested workers will provide optimal settings for running.
“Top features” and “Correlation |R| threshold” do not matter here.
3. Adjust the voxel dimensions (the pixel size in micrometres) to match your images.
4. Click “Run pipeline”.

When the run starts, the software checks the image naming but does not read labels.

The screenshot shows a web-based interface for a machine learning pipeline. On the left, a sidebar contains four main steps: 1. Generate features, 2. Train models, 3. Inference, and 4. All together. Step 1 is currently selected. The main panel on the right is divided into several sections. The top section, 'IMAGES TO CLASSIFY', has a button 'CHOOSE INFERENCE IMAGES' and a message 'No inference images uploaded.' Below this is a section for 'PARALLEL WORKERS' with a dropdown set to 4, and 'TOP FEATURES' with a dropdown set to 20. The 'CORRELATION |R| THRESHOLD' is set to 0.8. The 'Voxel dimensions (µm)' section has three dropdowns for X, Y, and Z dimensions, all set to 0.13, 0.13, and 0.5 respectively. A 'Suggested workers' section indicates 'Uses a conservative share of the CPUs available to this container.' Below that, a 'Where do the results go?' section explains that results appear in 'Inspect results' and can be downloaded as a ZIP file. At the bottom, a large green button labeled 'Run pipeline' is present, with a status message 'Ready for a new isolated job.' below it. Numbered callouts 1 through 5 are placed over the interface: 1 points to the 'Unlabelled' mode button, 2 points to the 'CHOOSE INFERENCE IMAGES' button, 3 points to the 'PARALLEL WORKERS' and 'TOP FEATURES' dropdowns, 4 points to the 'Voxel dimensions' dropdowns, and 5 points to the 'Run pipeline' button.

1 Generate features
Choose whether image filenames contain class labels.

☐ Labelled
For training images

☒ Unlabelled
For inference images

2 Train models
Upload a complete labelled feature table

3 Inference
Use uploaded images or a compatible feature table

4 All together
Generate, train, and classify in one isolated job

IMAGES TO CLASSIFY ?

CHOOSE INFERENCE IMAGES

No inference images uploaded.

PARALLEL WORKERS ? **TOP FEATURES** ?

4 20

CORRELATION |R| THRESHOLD ?

0.8

Voxel dimensions (µm) ?

X (µm) Y (µm) Z (µm)

0.13 0.13 0.5

Suggested workers Uses a conservative share of the CPUs available to this container.

Where do the results go?
They appear in **Inspect results** on this page. After the run, **Download results ZIP** saves a copy to your browser's configured download location; no output folder selection is needed.

Run pipeline

Ready for a new isolated job.

Do step (2) — run the model on those features:

1. Choose “Inference”.
2. Point the software to the folder containing inference images using “Choose inference images” **OR** point the software to the feature table (the features you just generated in unlabelled mode) using “Choose feature table”.
3. Point the software to the folder containing the trained model using the drop-down menu “Trained models”.
4. All trained models are saved in the Docker folder, so you need to use the drop-down menu and choose the appropriate model. They are saved by the date when they were trained, followed by the number of models trained (one, when only one model was selected, or six, when all learners were selected), followed by a code created by Docker for tracking in case of problems. If you would like to delete an old model, or a model you ran as a test, select the model from the drop-down menu and click on “Delete selected job”. You can also import trained models by using “Import _joblib files”. It takes you to the folder with models; mark those models you would like to import (all or just one). Once they are imported, they will be used for inference.
5. Set “Parallel workers” as before. Suggested workers will provide optimal settings for running.
6. Click “Run pipeline”.

The screenshot shows a web-based interface for a machine learning pipeline. On the left, a sidebar contains four main sections: '1 Generate features' (with 'Labelled' and 'Unlabelled' options), '2 Train models' (with an upload button), '3 Inference' (selected with a radio button), and '4 All together'. The main area on the right is divided into several sections: 'IMAGES TO CLASSIFY' with a 'CHOOSE INFERENCE IMAGES' button; 'COMPLETE FEATURE TABLE' with a 'CHOOSE FEATURE TABLE' button; 'TRAINED MODELS' with a dropdown menu and 'IMPORT _JOBLIB FILES' and 'Delete selected job' buttons; 'PARALLEL WORKERS' and 'TOP FEATURES' with input fields (4 and 10 respectively); 'CORRELATION |R| THRESHOLD' with an input field (0.8); 'Voxel dimensions (µm)' with fields for X (0.13), Y (0.13), and Z (0.5); 'Suggested workers' with a note about CPU usage; and 'Where do the results go?' with a note about result locations. At the bottom, a large green button labeled 'Run pipeline' is visible, with a note 'Ready for a new isolated job.' below it. Numbered callouts 1 through 5 are placed over the interface: 1 points to the 'Inference' section in the sidebar; 2 points to the 'CHOOSE INFERENCE IMAGES' and 'CHOOSE FEATURE TABLE' buttons; 3 points to the 'TRAINED MODELS' section; 4 points to the 'PARALLEL WORKERS' and 'TOP FEATURES' sections; and 5 points to the 'Run pipeline' button.

1 Generate features
Choose whether image filenames contain class labels.

☐ Labelled
For training images

☐ Unlabelled
For inference images

2 Train models
Upload a complete labelled feature table

3 Inference
Use uploaded images or a compatible feature table

4 All together
Generate, train, and classify in one isolated job

IMAGES TO CLASSIFY ?
CHOOSE INFERENCE IMAGES
No inference images uploaded.

COMPLETE FEATURE TABLE ?
CHOOSE FEATURE TABLE
No feature table uploaded.

TRAINED MODELS ?
Select a training job
Options show when the job was trained and which learner(s) it used. Import a compatible .joblib model below, or delete an old selected job. Persist /data with a Docker volume to keep jobs across container replacement.

IMPORT _JOBLIB FILES
Delete selected job
Select the model and matching metadata ".joblib" files together when available.

PARALLEL WORKERS ?
4

TOP FEATURES ?
10

CORRELATION |R| THRESHOLD ?
0.8

Voxel dimensions (µm) ?
X (µm) 0.13 Y (µm) 0.13 Z (µm) 0.5

Suggested workers Uses a conservative share of the CPUs available to this container.

Where do the results go?
They appear in **Inspect results** on this page. After the run, **Download results ZIP** saves a copy to your browser's configured download location; no output folder selection is needed.

Run pipeline
Ready for a new isolated job.